Team Selection Test Semester 1

Paly Math Club

2026-27

This test is the team-selection test for **Berkeley Math Tournament** and **Princeton University Mathematics Competition**.

Brief information on each competition:

BMT	PUMaC
November 14th, 2026	November 21st, 2026
Official	Unofficial
Location: Berkeley	Location: Princeton

Your decision on BMT is BINDING. Withdrawing on short notice may prohibit you from participating in future tournaments.

PUMaC is unofficial, so you will have to work it out with your teammates if you would like to join or withdraw on short notice.

Please circle the competitions you are planning to attend:

1) BMT 2) PUMaC

Instructions: This is a **60** minute test with **15** questions. Each question is given 1 point. No point deduction for incorrect answers, no partial credit except the estimation problem. You may only use a pe

Friendly Reminder: The perceived difficulty of problems varies from person to person. If you find one problem challenging, move on and try the others — you may find them more accessible.

Name: _____ Grade:_____

Email (list one actively checked): _____

Please list any relevant test scores/competition results (e.g. AMC, AIME).

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Please list any of your relatives that might be able to chaperone for competitions, including **which competitions** they would be available for and **their relation** to you (ex. BMT, John Doe, father). (This potentially may be used in team selection if we are severely lacking chaperones.)

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1 Algebra

1. If $x + \frac{1}{x} = 3$, what is $x^3 + \frac{1}{x^3}$?
2. If $(a+2)^{a-3} = (2a-5)^{a-3}$, find the sum of all possible real values of a.
3. Let P be the unique real polynomial of degree at most 6 such that $P(k) = k!$ for $k = 0, 1, 2, \ldots, 6$. Find $P(7)$.

2 Geometry

1. Triangle ABC has $AB = 4, BC = 5, CA = 6$. The tangent to the circumcircle of ABC at A meets line BC at P. What is the area of triangle PAB?
2. If a sphere is inscribed in a regular octahedron of side length 1, what is the sphere's surface area?
3. Let ABC be a triangle with side-lengths $5, 8, 12$. Find
$$\left((\sin A \cos A + \sin B \cos B + \sin C \cos C) \cdot \frac{1}{\sqrt{15}}\right)^{\frac{1}{2}}$$
4. Let α, β, γ be three positive real numbers such that they sum to $\frac{\pi}{2}$ and
$$4\cot(\alpha) = 4\cot(\beta) + 1 = 4\cot(\gamma) + 2$$
Find $\sin(2\alpha)$.

3 Combinatorics

1. How many ways can a positive number of kings be placed on a 3 by 3 chessboard such that no two kings attack each other? Note: the colors of the kings do not matter.
2. What is the number of 5-digit numbers that have strictly decreasing digits from left to right?
3. A frog starts at (0, 0) and must return to his home at (13, 7). There is a river located along the line $y = x - 6$. At each step, the frog can only move exactly one unit up or one unit to the right along the lattice points of the plane. If the frog cannot cross the river (but is allowed to move to points on the river), the number of paths the frog can take to his home is N. Compute $\frac{N}{120}$.

4 Number Theory

1. List the ordered positive integer pairs (a, b) that satisfy $2 \cdot \text{lcm}(a, b) + 3 \cdot \gcd(a, b) = 3a + 2b + 2026$?
2. Let
$$N = (3-1)(3^2-1)(3^3-1)\cdots(3^{10}-1)$$
What is the largest integer k such that 2^k divides N?
3. What is the largest prime factor of 280165? Some additional given information is that $\sqrt{280165} = |23^2 - 18i|$ and the answer is at least 169.
4. Let
$$f(n) = \sum_{d|n} \tau(d) \cdot \varphi\left(\frac{n}{d}\right)$$
where $\tau(n)$ is the number of divisors of n and $\varphi(n)$ is the number of positive integers less than or equal to n that are relatively prime to n. Find $f\left(67^{2026}\right) \pmod{1000}$.

5 Estimation

Problem: Estimate $A = \sum_{a=1}^{1000} \sum_{b=1}^{1000} \text{lcm}(a, b)$.

Scoring: Your score S is scaled to the interval $[0, 1]$ by $S = \min\left(\frac{E}{A}, \frac{A}{E}\right)^{15}$ where E is the value you submit and A is the exact value.